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Beyond Binary: Injury Level, Age, and Weaning Prediction in Cervical Spinal Cord Injury

Jiaxin Luo[1],Ying Zhu2*

1.The Fourth School of Clinical Medicine, Zhejiang Chinese Medical University, Hangzhou First People ' s Hospital, Hangzhou, Zhejiang Province,310000, China.

2.(Department of ICU)Affiliated Hangzhou First Ppeople ' s Hospital, ,School of Medicine,Westlake University ,Hangzhou Zhejiang Province 310000, China

*Corresponding author:doctorzhuying1977@163.com(YingZhu,Tel:15857187802);No. 261 Huansha Road, Shangcheng District, Hangzhou City, Zhejiang Province 310000,China

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Declaration

This autor takes responsibility for all aspects of the reliability and freedonm from bias of the presented and their discussed interpretation.

JL,YZ were major contributors in writing the manuscript. Both of the authors read and approved the final manuscript.

Ying Zhu is corresponding author.

Dear Editor,

We read with interest the retrospective analysis by Fenger and colleagues[1] examining ventilator weaning outcomes in 39 individuals with cervical spinal cord injury admitted to inpatient rehabilitation on continuous mechanical ventilation. The authors provide a useful granular description of weaning milestones—daytime wean by day 20, liberation from invasive ventilation by day 23, decannulation by day 36—and their finding that decannulation was strongly associated with discharge to home (OR 22.84) confirms what many clinicians suspect: removing the tracheostomy tube is a pivotal step toward community reintegration. I appreciated particularly their honesty about the timeline; weaning often takes weeks, and this information matters enormously when counselling families who arrive at rehabilitation exhausted and hoping for rapid progress.

That said, I found myself pausing at two points in the analysis, and I wonder whether the authors might consider these in future work.

The first concerns the categorisation of injury level. The authors divided patients into high (C1–C4) versus low (C5–C8) injuries, and this binary grouping drives much of their analysis. But in clinical practice, the distinction between C4 and C5 is not merely ordinal—it is physiological. C3, C4, and C5 innervate the diaphragm; patients with complete injuries below C5 retain phrenic innervation, whereas those with injuries at or above C4 may not. A recent retrospective study of 148 patients with traumatic cervical SCI found that after multivariate analysis, cord compression exceeding 25% and contusion length spanning at least two spinal levels were independent predictors of ventilator dependence, with outcomes differing substantially between C3–C4 and C5–C6 injuries[2]. Collapsing this spectrum into a binary risks obscuring what clinicians actually want to know: given this patient's specific level and completeness, what is the likely trajectory? The authors' own supplementary data might allow them to present outcomes stratified by finer groupings (C1–C2, C3–C4, C5–C6, C7–C8), and I suspect that would better serve clinicians trying to forecast outcomes for their patients.

The second point relates to the way discharge disposition was modelled. The authors found that age predicted discharge to more resource-intensive settings, with an odds ratio of 0.68 per decade. I agree that this is an important finding, but I wondered whether the analysis risked conflating age with something else. The authors did not report comorbidity burden, and it is plausible that older patients in their cohort had higher rates of pre-existing cardiac or pulmonary disease that independently influenced both weaning success and discharge options. A 2025 analysis of 71,048 patients with acute traumatic cervical SCI from the ACS Trauma Quality Programs database found that when frailty—measured by the modified Risk Analysis Index (mRAI)—was accounted for, age alone no longer independently predicted nonroutine discharge or in-hospital mortality[3]. Patients classified as "very frail" (mRAI ≥21) had more than double the odds of nonroutine discharge compared with robust patients, independent of age[3]. Without accounting for comorbidity or frailty, it is difficult to know whether age itself is driving the association or simply serving as a proxy for the accumulation of other risk factors. More critically, if older patients who achieve the

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same weaning milestones as younger patients are being discharged to different destinations, that raises questions about whether we are applying consistent criteria across age groups—or whether we are defaulting to institutional care for older patients even when they have the respiratory capacity to go home.

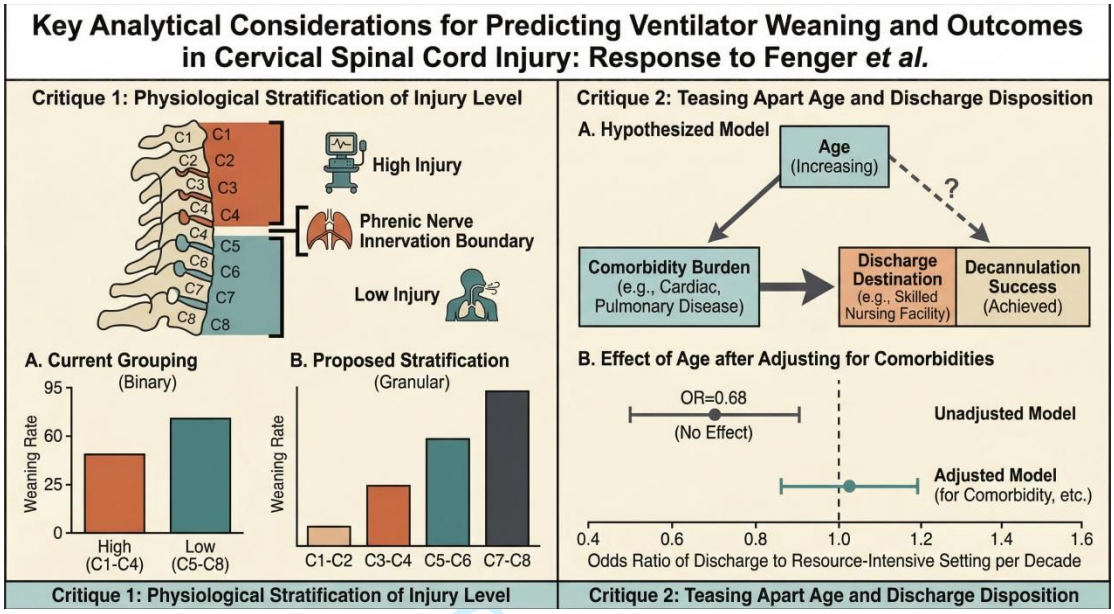

Figure1.Integrated Analytical Framework for Predicting Ventilator Weaning Outcomes in Cervical Spinal Cord Injury (CSCI).

The authors' dataset might allow them to test whether, among decannulated patients, age still predicted discharge destination after controlling for the factors they already collected. We raise these points not to diminish what the authors have accomplished,but because I believe the next step for this field is moving from “who succeeds” to "how do we predict success with enough precision to counsel patients and allocate rehabilitation resources wisely."

The authors' weaning protocol appears systematic and their outcomes enco uraging, particularly for high-level injuries where half achieved daytime wean. We would be curious whether, in their ongoing work, they have begun to inco rporate objective measures such as serial diaphragm ultrasound. A recent retros pective case series of 21 patients with cervical SCI found that diaphragm thick ening ratio (TR) on ultrasound was a strong predictor of weaning success: a th reshold of TR ≥ 1.2 had a sensitivity of 1.0 and specificity of 0.90 for predict ing successful liberation from mechanical ventilation[4]. Similarly, a 2025 qualit y improvement project implementing a multimodal weaning intervention—includ ing readiness-to-wean proforma, adapted RISCI guidelines, and weekly SCI war d rounds—reported that among patients with C2–C4 injuries, liberation from m echanical ventilation increased from 20% to 78% ($p = 0.032$) following the int ervention[5]. These data suggest that the gap between high and low-level injurie s may be narrower than previously assumed when structured, protocolised care is applied.

In the meantime, the authors' data serve as a useful benchmark for rehabil itation programmes seeking to evaluate their own performance, and I thank the m for the careful work.

References

[1] Fenger C, Locke B W, Barker J, et al. Postacute Ventilator Weaning and Discharge Outcomes in Individuals With Cervical Spinal Cord Injury: A Retrospective Analysis[J/OL]. Respiratory Care, 2026: 19433654261428406.

[2] Chaisawasthomrong C, Boongird A. Predictive Factors Affecting the Need for Mechanical Ventilation in Acute Traumatic Cervical Spinal Cord Injury.[J/OL]. Journal of neurotrauma, 2025, 42(23-24): 2178-2190.

[3] Elsamadicy A A, Serrato P, Belkasim S, et al. Assessing Morbidity and Mortality in Adult Patients With Acute Traumatic Cervical Spinal Cord Injury Using a Modified Risk Analysis Index.[J/OL]. Neurosurgery, 2025, 97(4): 829-841.

[4] Bhatia N S, Kunzweiler S, Conley C, et al. Predictive Value of Diaphragm Muscle Ultrasound for Ventilator Weaning Outcomes After Cervical Spinal Cord Injury: A Retrospective Case Series.[J/OL]. Journal of ultrasound in medicine : official journal of the American Institute of Ultrasound in Medicine, 2025, 44(1): 119-126.

[5] Lea T J, Harriman A J C, Hodson J, et al. A multimodal respiratory weaning intervention for adult critical care patients mechanically ventilated with cervical spinal cord injury: a quality improvement project.[J/OL]. Spinal cord, 2025, 63(11): 579-587.

Figure Legend: Integrated Analytical Framework for Predicting Ventilator Weaning Outcomes in Cervical Spinal Cord Injury (CSCI).

(Left Panel) Physiological stratification by spinal level: Traditional binary grouping (High vs. Low) is refined into a granular continuum (C1–C2, C3–C4, C5–C6, C7–C8) based on phrenic nerve innervation boundaries, demonstrating improved weaning rate stratification.

(Right Panel) Causal pathway dissection: The hypothesized model (A) illus trates the confounding effects of age and comorbidity burden on discharge disp osition. After adjusting for comorbidities (B), the independent effect of age on discharge to resource-intensive settings is minimized (Adjusted Model), highligh ting the need for equitable, function-based discharge planning rather than age-b ased stratification alone.

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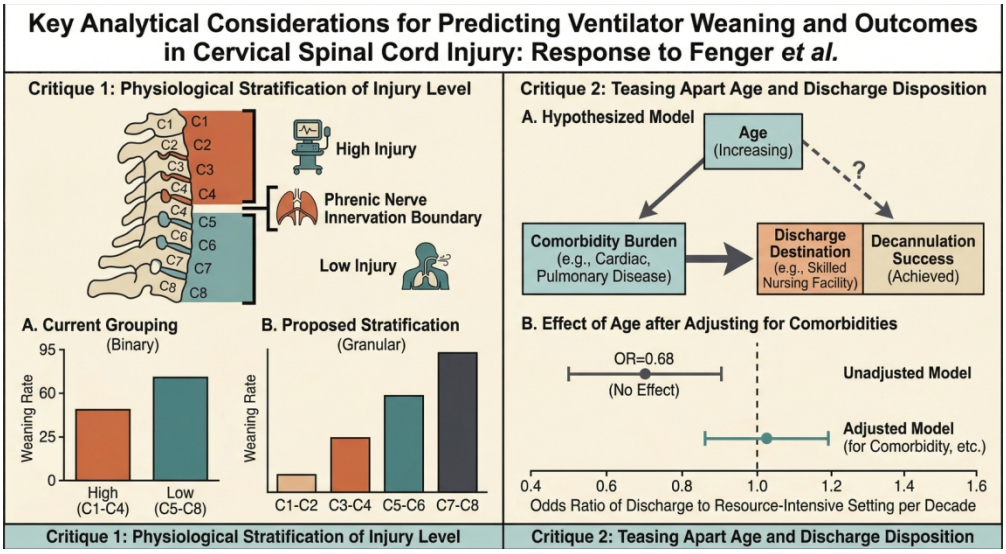

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(Right Panel) Causal pathway dissection: The hypothesized model (A) illustrates the confounding effects of age and comorbidity burden on discharge disposition. After adjusting for comorbidities (B), the independent effect of age on discharge to resource-intensive settings is minimized (Adjusted Model), highlighting the need for equitable, function-based discharge planning rather than age-based stratification alone.

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